

Jaskirat Singh | Academic CV

3rd year Ph.D. Student in Computer Science | Intern @ Meta GenAI

✉ Email: jaskirat.singh@anu.edu.au 🌐 Homepage 🎓 Google Scholar

Research Interests

Video Generation, Controllable Image Synthesis and Editing, Creative Content Generation

Professional Experience

Research Scientist Intern

Imagine-Core (Video Generation) with [Prof. Michael Cohen](#)

Meta GenAI

May' 24 – Nov' 24

Research Scientist Intern

Creative Intelligence Lab with [Dr. Zhe Lin](#) & [Dr. Jianming Zhang](#)

Adobe Research

Jun' 23 – Sep' 23

Research Assistant

Geometry of Learning Lab with [Prof. Richard Hartley](#)

Australian National University

Mar' 23 – Jun' 23

Research Scientist Intern

Graphics Intelligence and Learning Lab with [Dr. Jose Echevarria](#)

Adobe Research

Jun' 21 – Dec' 21

Machine Learning Research Engineer

Mathematical Modeling for Optimizing Generative Image Content & Design

Yahoo! Japan

Oct' 17– Sept '18

Education

◦ The Australian National University

Ph.D. in Computer Science

Supervisors: [Prof. Liang Zheng](#) and [Prof. Stephen Gould](#)

Expected

Sep' 21–Present

◦ The Australian National University

Master of Machine Learning and Computer Vision

🎓 Awarded University Medal for Exceptional Academic Excellence

GPA: 7/7

Jul' 19–Jul' 21

◦ Indian Institute of Technology, Delhi

Bachelor of Technology (B.Tech), Electrical Engineering

Specialization in Intelligent and Cognitive systems

GPA: 9.3/10

2013–2017

Selected Publications

1. Negative Token Merging: Image-based Adversarial Feature Guidance

Jaskirat Singh, Lindsey Li, Weijia Shi, Ranjay Krishna, Yejin Choi, Pang Wei Koh, Michael F. Cohen, Stephen Gould, Liang Zheng, Luke Zettlemoyer.

2024

🏆 Image based classifier-free guidance to improve diversity, quality in few lines of code

2. SmartMask: Context Aware High-Fidelity Mask Generation for Fine-grained Object Insertion and Layout Control

Jaskirat Singh, Jianming Zhang, Qing Liu, Cameron Smith, Zhe Lin, Liang Zheng. CVPR 2024, US Research Patent (Adobe)

2024

3. OpenDevin: An Open Platform for AI Software Developers as Generalist Agents

Best LLM Agent for 🏆 SWE-Bench Verified & 🏆 SWE-Bench-Lite.

2024

ICLR 2025

4. [IMPUS: Image Morphing with Perceptually-Uniform Sampling Using Diffusion Models](#)
Zhaoulun Yang, Zhengyang Yu, **Jaskirat Singh**, Zhiwei Xu, Jing Zhang,
Dylan Campbell, Peter Tu, and [Richard Hartley](#). 2023
ICLR 2024
5. [Divide, Evaluate, and Conquer: A Compositional Perspective on Evaluating and Improving Text-to-Image Alignment](#)
Jaskirat Singh, and [Liang Zheng](#). 2023
NeurIPS 2023
6. [High-Fidelity Guided Image Synthesis with Latent Diffusion Models](#)
Jaskirat Singh, [Stephen Gould](#), and [Liang Zheng](#). 2023
CVPR 2023
7. [Paint2Pix: Interactive Painting based Progressive Image Synthesis and Editing](#)
Jaskirat Singh, [Liang Zheng](#), Cameron Smith, and Jose Echevarria. 2022
ECCV 2022
8. [Intelli-Paint: Towards Developing Human-like Painting Agents](#)
Jaskirat Singh, Cameron Smith, [Jose Echevarria](#), and [Liang Zheng](#). 2022
ECCV 2022, US Research Patent (Adobe)
9. [Combining Semantic Guidance and Deep Reinforcement Learning For Generating Human-Level Paintings](#)
Jaskirat Singh, and [Liang Zheng](#). 2021
CVPR 2021

Honors and Achievements

- 🏆 **Awarded University Medal** for academic excellence at the Australian National University.
- 🏆 **Awarded Chancellors Letter of Commendation** at the Australian National University.
- Awarded ANU Computer Science Summer Research Grant (\$5k).
- **Invited for delivering a tutorial** on "Applying deep reinforcement learning for computer vision research" by the **Australian Centre for Robotic Vision (ACRV)** group.
- Our project "Connected Stories of Australia" has been awarded as the **best innovative design project** by the **National Museum of Australia**.
- **Won national hackday at Yahoo Japan**, among 54 competing teams from all across Japan, for developing a real-time application for **facial attribute modification using reversible GANs**.
- Received **IIT Delhi Merit Award & Scholarship** for outstanding academic performance.
- Secured **All India Rank 128 in IIT-JEE** among 1.4 million aspirants appearing for the exam.
- 🏆 Won the **Silver Medal at National FIDE Rated Chess Tournament**.

Referees

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| ◦ Prof. Liang Zheng - Associate Professor | 🏛️ Australian National University |
| ◦ Prof. Stephen Gould - Senior Professor | 🏛️ ANU, Amazon Research |
| ◦ Prof. Marcus Hutter - Senior Research Scientist | 🏛️ Google Deepmind |
| ◦ Dr. Jianming Zhang - Principal Research Scientist | 🏛️ Adobe Research |
| ◦ Dr. Zhe Lin - Senior Principal Research Scientist | 🏛️ Adobe Research |
| ◦ Prof. Michael Cohen - Director of Computational Photography | 🏛️ Meta GenAI, UW |